SMART AGRICULTURE DECISION SUPPORT SYSTEM FOR SRI LANKAN PADDY CULTIVATION

GROUP RESEARCH PROJECT

R26-IT-066

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B.Sc. (Hons) Degree in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

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DECLARATION OF THE CANDIDATES AND SUPERVISOR

We declare that this group dissertation is based on the research and implementation work undertaken under Project R26-IT-066. Material derived from published sources and the individual component reports is acknowledged through citations and references. Each member remains responsible for the accuracy and originality of the work attributed to their component. The integrated report preserves the distinction between implemented functionality, measured results, design intentions and work that remains to be validated.

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ABSTRACT

Paddy cultivation is central to food security and rural livelihoods in Sri Lanka, yet farmers make decisions under changing field conditions, nutrient variability, disease pressure, uncertainty in expected harvest and fluctuating market prices. Digital agricultural tools often address these concerns separately. Project R26-IT-066 develops a modular Smart Agriculture Decision Support System intended to connect four complementary capabilities: Internet of Things (IoT)-based field monitoring and fertilization guidance; crop recommendation with market-price analysis; paddy leaf disease detection and severity analysis; and disease-aware harvest-yield estimation with treatment decision support.

The project adopts an engineering and design-science-oriented approach in which each student develops a distinct component and evaluates it using the available model, software and prototype evidence. The field-monitoring component uses an ESP32-based station, Firebase Realtime Database, a Flutter interface and a Flask prediction service. Its reported fertilization classifier achieved 0.9400 accuracy on a held-out set of 4,800 records from a 24,000-record dataset, while the fertilizer-amount regressor reported $R^2 = 0.8666$. The disease-detection component uses YOLOv8 for annotated paddy-leaf image analysis. Its reported plots show approximately 0.950 mAP@0.5 across displayed classes, although class imbalance and inconsistencies in class naming require further review. The crop-recommendation component combines soil and climate inputs with season and price-related inputs, but the reported comparison of Random Forest, XGBoost and Decision Tree achieved only approximately 0.33, 0.44 and 0.39 accuracy respectively. The harvest component separates baseline-yield prediction using Random Forest from disease-loss estimation using XGBoost and computes untreated and treated scenarios; however, the treatment scenario currently assumes a 60% loss reduction, which is not validated as an agronomic effect.

The combined evidence supports the existence of several implemented prototypes and component-level analytical workflows. It does not yet establish that the entire system has passed end-to-end integration testing or is ready for autonomous farmer-

facing decisions. Important remaining work includes harmonising input and output contracts, validating cross-component data exchange, resolving dataset and label limitations, recording final model metrics, verifying API behaviour, measuring latency and usability, and evaluating the treatment and economic assumptions in field conditions. The principal contribution is a modular architecture that connects field observations, crop and disease analysis, harvest estimation and decision support while explicitly documenting the evidence and limitations of each component.

Keywords: smart agriculture; paddy cultivation; Internet of Things; fertilization prediction; crop recommendation; market price analysis; YOLOv8; disease severity; yield estimation; decision support system.

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The team acknowledges the Department of Information Technology at the Sri Lanka Institute of Information Technology for providing the academic environment in which the project was conducted. The members also acknowledge the cooperation among the four individual components. The IoT field-monitoring component provides field-condition data; the crop recommendation and market component explores variety selection; the disease component analyses paddy-leaf images; and the harvest-yield component translates field and disease information into harvest-oriented scenarios.

Finally, the team thanks family members, friends and all others who provided encouragement during the design, implementation, testing and documentation stages.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
DSS	Decision Support System
DoA	Department of Agriculture
DSR	Design Science Research
ESP32	Low-cost microcontroller family used for connected sensing
IoT	Internet of Things
MAE	Mean Absolute Error
mAP	Mean Average Precision
ML	Machine Learning
NPK	Nitrogen, Phosphorus and Potassium
RF	Random Forest
RMSE	Root Mean Squared Error
R^2	Coefficient of determination
REST	Representational State Transfer
SLIIT	Sri Lanka Institute of Information Technology
UI	User Interface
YOLO	You Only Look Once
XGBoost	Extreme Gradient Boosting
Firebase	Cloud platform used in the project for realtime data storage and synchronisation

1. INTRODUCTION

1.1 Background and Context

Agriculture is increasingly supported by connected sensors, machine-learning models, computer vision and mobile applications. These technologies can help farmers observe field conditions, interpret crop symptoms and compare management choices. Their practical value depends not only on model performance but also on the quality of data, reliability of communication, clarity of outputs and relevance to local agricultural practice.

Paddy cultivation is especially suitable for a connected decision-support approach because field conditions, nutrient availability, crop variety, disease pressure and harvest outcomes are interdependent. Soil and environmental measurements can inform fertilization decisions. Crop and seasonal information can support variety selection. Images of paddy leaves can be analysed to identify visible disease patterns. Finally, yield estimates and disease-related losses can be translated into harvest scenarios. When these capabilities are disconnected, farmers may receive isolated outputs without a consolidated interpretation.

Project R26-IT-066 responds to this problem by developing four complementary components. The intended system is modular rather than monolithic: each component has a defined responsibility and its own data, model and user-interface requirements. The group-level objective is to establish how these components can form a coherent smart-agriculture decision-support workflow for Sri Lankan paddy cultivation.

1.2 Overall Project Context

The project comprises the following components, each documented in an individual thesis and brought together in this group report.

Student ID	Component	Main technical elements
IT22178718	Real-time IoT field monitoring and fertilization prediction	ESP32-based sensing, Firebase Realtime Database, Flutter field screen, Flask service, Random Forest classifier and XGBoost

		regressor
IT21811104	Crop recommendation and market analysis	Flutter Smart Crop Guide, Flask prediction service, Random Forest crop model, market-price model and variety ranking
IT22261632	Paddy disease detection and severity analysis	Paddy leaf images, image preparation and annotation, YOLOv8 object detection, disease-region and severity workflow
IT22266446	Disease-aware harvest yield estimation and treatment decision support	Random Forest baseline-yield model, XGBoost disease-loss model, scenario calculations, Flask endpoint and Flutter interface

1.3 Problem Statement

The overarching research problem is that agricultural decision-making often relies on fragmented observations and separate digital tools. A field-monitoring system may report nutrient readings without incorporating disease symptoms. A disease classifier may identify a leaf condition without estimating its possible effect on harvest. A yield model may estimate production without representing disease-related loss. A variety recommendation may consider soil and climate but not robustly integrate market price and expected revenue. These gaps make it difficult to transform separate predictions into an interpretable decision-support process.

The group research therefore investigates how a modular system can collect or accept relevant field and crop information, analyse disease images, support fertilization and variety selection, and estimate harvest outcomes while documenting the reliability and limitations of its outputs. The project does not assume that all planned functions are already fully integrated or validated. The current state of each component is assessed against the evidence available in its individual report.

1.4 Research Gap

The combined component reports identify four related gaps. First, field measurements need to be translated into actionable fertilization guidance rather than remaining as raw sensor values. Second, crop recommendation needs to consider a wider set of factors, including season, harvest timing and market information, while

being supported by reliable model evaluation. Third, disease identification should go beyond a simple disease label where possible by detecting regions and estimating severity. Fourth, disease information needs to be connected to expected harvest impact so that farmers can compare untreated and treated scenarios.

Gap area	Observed limitation	Project response	Current qualification
Field monitoring	Raw sensor readings alone do not tell the farmer whether fertilization is needed or in what amount.	ESP32 telemetry, Firebase, Flutter and a two-stage fertilization model.	Reported model metrics are promising, but growth-stage encoding and recommendation logic need correction.
Crop and market decisions	Variety choice may not account for seasonal and market uncertainty; model evidence is weak.	Crop recommendation prototype with a market-price model and ranked output.	Classifier accuracy is low; price-model metrics and ranking validation are missing.
Disease analysis	A disease name alone does not describe the visible location or extent of symptoms.	YOLOv8-based paddy-leaf object detection and severity workflow.	Class imbalance, naming consistency and independent validation need review.
Harvest decision support	Disease labels do not directly estimate expected yield loss or compare treatment scenarios.	Separate baseline-yield and disease-loss models with scenario calculations.	Treatment efficacy is assumed at 60%; final model and end-to-end validation evidence is incomplete.
System-level integration	Individual outputs may be produced by separate services and data contracts.	Modular architecture intended to connect the four components.	A complete end-to-end integration test is not established by the supplied evidence.

1.5 Main Objective

To design, implement and evaluate a modular Smart Agriculture Decision Support System for Sri Lankan paddy cultivation that combines IoT-based field monitoring and fertilization guidance, crop recommendation with market analysis, paddy disease detection and severity analysis, and disease-aware harvest-yield estimation to support informed agricultural decisions.

1.6 Specific Objectives

1. To acquire and present field-condition and nutrient readings through an IoT-oriented monitoring workflow and evaluate a model-based fertilization decision mechanism.

2. To develop a crop recommendation prototype that accepts field, seasonal and market-related inputs and returns ranked paddy-variety suggestions, while evaluating the reliability of the underlying models.
3. To develop a computer-vision workflow for identifying annotated paddy-leaf disease patterns using YOLOv8 and to examine the available detection and class-level results.
4. To develop a disease-aware yield-estimation workflow that separates baseline yield from disease-related loss and presents untreated and treated scenarios.
5. To define the interfaces and data flows required for the four components to operate as parts of one decision-support platform.
6. To evaluate the available model, functional, API and integration evidence; identify limitations; and propose realistic next steps for validation and deployment.

1.7 Research Questions

7. How can field sensor data be used to support fertilization decisions in a paddy cultivation context?
8. To what extent does the current crop-recommendation prototype produce reliable variety rankings from field and market-related inputs?
9. What evidence is available for YOLOv8-based paddy-leaf disease detection, and what limitations affect its generalisation?
10. How can baseline-yield and disease-loss estimates be combined into understandable harvest scenarios?
11. What interface, data-quality and validation conditions must be satisfied before the four prototypes can be treated as a dependable integrated decision-support system?

1.8 Scope and Limitations

The scope is limited to the four implemented or prototyped components described in the submitted individual reports. It includes their stated data inputs, model artefacts, service and interface layers, and the results reported in those documents. It does not

claim that all components are deployed in a production environment, that all outputs are agronomically validated, or that the integrated system has completed a field trial.

The evidence has several limitations. The crop-recommendation component reports low classifier accuracy and lacks documented market-price model metrics. The IoT component reports strong classifier performance but identifies a growth-stage encoder mismatch and a latest-reading-only telemetry store. The disease component reports promising detection metrics on its displayed evaluation, but the dataset is imbalanced and class labels need harmonisation. The yield component lacks a verified final experiment log for several metrics and uses an unvalidated 60% treatment-effect assumption. These limitations are carried forward into the results and discussion rather than being concealed.

1.9 Individual Contributions

Student ID	**Individual contribution represented in this report**
IT22178718	Field monitoring and fertilization guidance, including sensor/telemetry workflow, classifier and amount regressor.
IT21811104	Crop recommendation and market-price analysis, including candidate variety ranking and baseline model comparison.
IT22261632	Paddy leaf disease detection and severity-analysis workflow using YOLOv8 and annotated image data.
IT22266446	Baseline-yield prediction, disease-loss estimation and scenario-based harvest decision support.

1.10 Structure of the Dissertation

Chapter 1 introduces the overall problem, gap, objectives and scope. Chapter 2 reviews the technical foundations and related work represented in the component reports. Chapter 3 explains the research methodology and evaluation strategy. Chapter 4 describes the architecture, implementation and intended integration of the four components. Chapter 5 consolidates the available results and validation evidence. Chapter 6 discusses the findings and limitations. Chapter 7 addresses

commercialisation, ethics and sustainability. Chapter 8 concludes the report and identifies future work.

2. LITERATURE REVIEW AND RELATED WORK

2.1 Review Scope

The literature review is organised around the technologies and research tasks used in the project: IoT-based field monitoring, machine learning for fertilization and yield estimation, crop recommendation and market-aware decision support, computer vision for plant disease detection, and integrated agricultural decision-support systems. The submitted individual reports include their own references and literature discussions; this group synthesis retains the central technical themes while avoiding claims that cannot be supported by the supplied documents.

2.2 Smart Agriculture and Decision Support

Smart agriculture combines sensing, communication, data processing and decision-support functions. Sensors can capture environmental or soil conditions, but measurements only become useful when they are interpreted against an agricultural task. A decision-support platform should therefore make clear which inputs are measured, which values are inferred by a model, what action is suggested, and what uncertainty remains. In the present project, the field-monitoring module provides readings, while fertilization, crop recommendation and yield modules transform selected inputs into predictions or scenarios.

The modular architecture is valuable because different tasks require different algorithms and validation methods. Sensor telemetry should be evaluated for data completeness and update behaviour. Classification and regression models require appropriate held-out testing and transparent metrics. Image detection requires class-wise evaluation and inspection of annotation quality. Scenario calculations require mathematical consistency and domain validation. These forms of evidence are complementary but not interchangeable.

2.3 Internet of Things for Field Monitoring

IoT-based agriculture uses connected sensing devices to collect and transmit field observations. The IT22178718 component describes an ESP32-based station that

publishes moisture, temperature, humidity and nutrient readings to Firebase Realtime Database. A Flutter field screen synchronises readings, while a Flask prediction service provides fertilization decisions. This architecture separates physical sensing from application presentation and model inference.

A key benefit of this design is the ability to use recent field observations rather than relying exclusively on manual entry. However, a sensor-driven workflow depends on sensor calibration, network availability, timestamping, missing-value handling and correct interpretation of the most recent observation. The individual report notes that the telemetry store keeps only the latest reading. That design may be sufficient for a current-value screen, but it limits historical trend analysis unless a separate history mechanism is added.

2.4 Machine Learning for Fertilization and Yield

Random Forest and XGBoost are used in multiple components. Random Forest aggregates predictions from decision trees and can represent nonlinear relationships among structured variables. XGBoost builds boosted tree ensembles and is used here for regression tasks such as fertilizer amount or disease-related loss. Model choice alone does not establish suitability: performance depends on the data, label construction, feature representation, split strategy and evaluation metrics.

The fertilization component uses a two-stage process: a Random Forest classifier decides whether fertilizer is needed and an XGBoost regressor estimates an amount in kg/ha. Its reported held-out results are 0.9400 accuracy for the classifier and R^2 = 0.8666 for the amount regressor. However, the report identifies that nitrogen, phosphorus and potassium dominate the classifier's feature importance, accounting for 97.2% collectively, and that some growth-stage values sent by the mobile interface are rejected by the model encoder. These findings demonstrate why data and interface consistency are essential to the interpretation of apparently strong metrics.

The yield component separates baseline yield from disease-related loss. The Random Forest baseline model receives environmental, nutrient and crop variables. The XGBoost loss model receives disease-related variables such as disease type, severity

and growth stage. The service then calculates untreated and treated scenarios. Separating these stages can make the system easier to interpret, but the final scenario remains only as reliable as its models and assumptions. In particular, a fixed treatment-effect assumption should not be presented as an observed agronomic outcome.

2.5 Crop Recommendation and Market-Aware Decisions

Crop recommendation commonly maps soil and environmental features to a crop or variety class. A market-aware approach additionally considers season, harvest timing and market price so that the result can potentially inform economic viability. The IT21811104 component implements a Smart Crop Guide interface and a Flask service. The reported service predicts prices for candidate varieties, applies fixed variety bonuses and returns the top three ranked varieties.

The model-comparison results reported in the individual thesis are approximately 0.44 accuracy for XGBoost, 0.39 for Decision Tree and 0.33 for Random Forest. These results are well below the proposal target of 85%, and the deployed model is not the highest-scoring of the three. The report also notes a repeated yield value of 2800 in roughly half the dataset, a very small evaluation set, an undocumented classification target and absent price-model metrics. Therefore, the component is treated as a prototype and baseline rather than a reliable source of farmer recommendations.

2.6 Computer Vision for Paddy Disease Detection

Computer vision can help identify visible disease patterns from leaf images. The IT22261632 component uses YOLOv8 to detect annotated regions in paddy-leaf images and includes a workflow for image processing, disease detection and severity assessment. The reported dataset summary contains 957 images or annotated instances across six displayed classes: Rice Blast, Brown Spot, Healthy, Leaf Blast, Narrow Brown and Leaf Scald. The individual report also discusses disease categories including bacterial disease. This variation in terminology should be reconciled before the final dataset and class taxonomy are frozen.

The displayed training evidence reports approximately 0.950 mAP@0.5, precision around 0.92 and recall around 0.93 from the plotted curves. Per-class AP values shown in the report include 0.995 for Bacterial, 0.904 for Brown Spot, 0.974 for Healthy, 0.885 for Leaf Blast and 0.993 for Narrow Brown. These are reported from the supplied plots and should be verified against the final evaluation output. Leaf Scald has only 23 examples in the dataset summary, which may limit reliable generalisation for that class.

Severity analysis introduces additional uncertainty because the detected region must be translated into a meaningful estimate of affected area or severity category. The relationship between image-derived area and agronomic severity should be clearly defined and validated. The system should not imply that a bounding-box output automatically equals a clinically or agronomically confirmed disease diagnosis.

2.7 Related Work and Comparative Perspective

Technology area	Strength of approach	Key caution for this project
IoT field monitoring	Provides timely measurements and can reduce dependence on manual entry.	Sensor calibration, network failures, history retention and timestamp alignment require testing.
Tree-based fertilization models	Can learn nonlinear relationships and support structured inputs.	High accuracy can reflect label rules or dominant features rather than general agronomic validity.
Market-aware crop recommendation	Can combine field suitability and economic signals in one interface.	Market training data, temporal validation and net-revenue calculations are not fully evidenced.
YOLO-based disease detection	Can localise visible regions and evaluate detection with precision, recall and mAP.	Class imbalance, taxonomy consistency, image quality and external validation remain important.
Disease-aware yield estimation	Separates baseline yield from disease-related losses and presents scenarios.	Treatment-effect assumptions and loss-model metrics must be empirically justified.
Integrated decision support	Can combine separate outputs into a farmer-facing workflow.	Integration must be verified with consistent data contracts and end-to-end tests.

2.8 Research Gap Synthesis

The shared gap is not simply the absence of a single algorithm. It is the lack of a verified end-to-end process that turns field observations and crop symptoms into

transparent, validated agricultural decisions. The four components address different stages of that process, but their usefulness depends on consistent data interfaces, reliable model evaluation, domain-aware interpretation and a clear presentation of uncertainty. This project contributes a modular prototype architecture and component-level evidence; full system-level validation remains future work.

2.9 Chapter Summary

This chapter reviewed the technical themes represented in the four components. The main implication is that each technology should be evaluated according to its role: telemetry quality for IoT, predictive performance and labels for machine learning, class-wise metrics for disease detection, and consistency of assumptions for yield scenarios. These considerations guide the methodology and results chapters.

3. RESEARCH METHODOLOGY

3.1 Research Design

The project follows an applied software-engineering and design-science-oriented approach. The team identifies an agricultural decision problem, designs a technical artefact, implements component workflows and evaluates the artefacts using available datasets, model outputs, screenshots, API behaviour and functional scenarios. The four components are developed separately because their inputs and analytical tasks differ, then documented under a shared system objective.

The methodology is iterative. Requirements inform architecture and implementation; implementation exposes data and integration constraints; evaluation identifies weaknesses; and those findings inform the final recommendations. The group report does not combine the component metrics into a single score because the tasks and evaluation measures differ.

3.2 Research Phases

Phase	Activities	Expected evidence
1. Problem definition	Identify agricultural decisions that need support and review the four individual research scopes.	Problem statement, gap synthesis and objectives.
2. Requirement analysis	Define inputs, outputs, users, data dependencies and constraints for each component.	Functional and non-functional requirements.
3. Data preparation	Prepare structured field datasets and annotated paddy-leaf images; inspect feature labels and class distributions.	Dataset summaries, preprocessing records and annotation evidence.
4. Model development	Train or load the component-specific Random Forest, XGBoost, decision-tree and YOLOv8 artefacts.	Model files, training plots and metric outputs.
5. Application implementation	Implement Flutter interfaces, Flask services, Firebase connection and component workflows.	Screenshots, source-code evidence and API outputs.
6. Component evaluation	Evaluate model metrics, behavioural scenarios, functional behaviour and output consistency.	Confusion matrices, regression metrics, scenario tests and test logs.
7. Integration review	Inspect dependencies, payload	Interface map and end-to-end

	fields, service routes and expected data flows across components.	test plan.
8. Reporting	Consolidate results, limitations, contributions and future work.	This group dissertation and evidence checklist.

3.3 Data and Study Materials

The study materials are the datasets, model artefacts, application code, service definitions, screenshots, training plots and test records described in the individual reports. The group report does not claim a new shared dataset spanning all four components. Each model operates on the data structure appropriate to its task.

Component	Inputs / study material	Output or decision
IoT and fertilization	Moisture, temperature, humidity, N, P, K and encoded crop/context fields; 24,000-record field dataset as reported.	Whether fertilizer is needed and estimated amount in kg/ha.
Crop recommendation and market	Soil and climate features, cultivation season, planned harvest month and current market price; ten-column paddy dataset as reported.	Ranked paddy-variety recommendations with predicted price information.
Disease detection	Annotated paddy-leaf images; 957 examples/instances across six displayed classes in the individual report.	Detected disease-related regions and associated class/confidence outputs.
Yield and treatment support	Field and crop-condition variables plus disease type, severity and growth stage; stored model bundles.	Baseline yield, disease-loss estimate, untreated and treated scenario values.

3.4 Data Preparation and Feature Handling

Data preparation is component-specific. Structured datasets require checking column names, data types, missing values, target definitions, class balance and train/test separation. Image datasets require consistent class names, valid annotations and a clear separation between training, validation and test images. In both cases, preprocessing decisions must be documented so that reported results can be reproduced.

The IoT component's report describes a 24,000-record dataset and a 20% held-out test set, equivalent to 4,800 test records. It also identifies that the classifier label

follows a nutrient-threshold rule and that NPK features dominate the model. This matters because high performance can reflect the structure of the labels rather than the full complexity of fertilizer need in real fields.

The crop-recommendation component reports a ten-column dataset, with approximately half of yield values equal to 2800 and an undocumented classification target. Before final submission or deployment, the data origin, duplicate patterns, target derivation and evaluation sample size should be verified. No additional dataset size or accuracy should be inferred where the individual report does not provide it.

The disease component's dataset summary reports 957 examples or annotated instances across six classes, including a small Leaf Scald class with 23 examples. The final report should clarify whether the count refers to images, annotations or class instances, and align the names in the dataset, training configuration, evaluation plots and application output.

The yield component stores model bundles with feature-column expectations. The feature order used at training and inference must remain consistent. The individual report also describes a mismatch between fields expected by the backend and the Flutter request payload, so data-contract verification is required before claiming end-to-end integration.

3.5 Model Development and Evaluation

3.5.1 Fertilization prediction

The fertilization workflow uses a Random Forest classifier to predict whether fertilizer is needed and an XGBoost regressor to estimate the required amount. The classifier is evaluated with accuracy, precision, recall, F1 and confusion-matrix information. The regressor is evaluated using R^2 and should also be accompanied by MAE and RMSE when the final verified outputs are available. The component report records classifier accuracy of 0.9400 on 4,800 held-out records and R^2 of 0.8666 for the regressor.

3.5.2 Crop recommendation and market analysis

The crop component compares Random Forest, XGBoost and Decision Tree classifiers. Its current comparison is reported as accuracy only, with approximate

scores of 0.33, 0.44 and 0.39 respectively. A complete evaluation should include a confusion matrix, per-class precision and recall, F1-score, cross-validation and a documented target. The market-price model needs a defined training dataset and regression metrics such as MAE, RMSE and R^2. These missing metrics are not estimated in this group report.

3.5.3 Paddy disease detection

The disease component uses YOLOv8 with 640 × 640 image input, batch size 16 and approximately 50 training epochs as described in the individual report. The report includes loss curves, precision and recall curves, mAP and per-class AP values. For a reproducible final evaluation, the team should state the exact model variant, data split, hardware/software environment, confidence threshold, IoU threshold and final exported metrics. Approximate values read from plots should be labelled as approximate.

3.5.4 Yield and disease-loss estimation

The yield component uses Random Forest regression for baseline yield and XGBoost regression for disease-related loss. The service derives scenario outputs from these estimates. Its individual report states that the treated scenario assumes a 60% reduction in predicted disease loss. This is a software assumption, not a validated treatment efficacy. Final evaluation must include the verified test-set size and held-out R^2, MAE and RMSE for each regression task, together with scenario arithmetic checks.

3.6 Functional and Non-Functional Evaluation

Functional evaluation asks whether the component accepts valid input, rejects invalid input appropriately, returns the expected output structure and displays results correctly. Non-functional evaluation considers latency, reliability, usability, data security, maintainability and availability. The supplied reports contain evidence at different levels; this group report distinguishes implementation evidence from completed verification.

Evaluation area	Minimum test or evidence required
Input validation	Valid, missing, out-of-range and incorrectly typed

	inputs for every component.
Model inference	Known test records, expected output shape, model version and reproducible metrics.
API behaviour	Successful request/response, HTTP status, error handling and response schema.
Scenario arithmetic	Check baseline, loss, untreated and treated values for mathematical consistency.
Data synchronisation	Confirm timestamps, current readings, missing data and Firebase/network failure behaviour.
Integration	Verify that output fields from one component match the expected input fields of the next component.
Latency and reliability	Record repeated response times, failures and recovery behaviour under defined conditions.
Usability	Document participant profile, tasks, method, consent, scores and qualitative feedback if a study is conducted.

3.7 Ethical, Privacy and Responsible AI Considerations

The system concerns agricultural decisions and can influence the use of fertilizer, treatment products and crop planning. Predictions should therefore be presented as decision support rather than guaranteed agronomic advice. Model limitations, uncertainty and the distinction between observed measurements and inferred outputs should be visible to users.

Data governance should include the source and permitted use of datasets, protection of any identifiable farmer or farm information, secure handling of credentials and API endpoints, and retention policies for telemetry. Image datasets should be used according to their licensing and collection conditions. Where farmer interviews or field studies are conducted, informed consent and appropriate institutional procedures should be followed. No unreported farmer usability study is claimed in this group report.

3.8 Reproducibility

Reproducibility requires storing the code version, model files, training configuration, dataset version, feature schema, random seeds, evaluation scripts and exact output logs. Screenshots are useful supplementary evidence but should not replace raw metric outputs or test records. The final submission should identify which metrics

come from saved training logs, which were calculated independently and which are approximate readings from plots.

3.9 Chapter Summary

The methodology treats each component according to its analytical role while using shared principles for data quality, model evaluation, interface verification and responsible reporting. The following chapter describes the intended architecture and the current implementation boundaries.

4. SYSTEM DESIGN, IMPLEMENTATION AND INTEGRATION

4.1 Design Principles

The system is designed as four complementary modules rather than one model with a single output. This separation allows each student to develop a specialised function while preserving possible integration points. The design principles are modularity, explicit input/output contracts, separation of model inference from user-interface presentation, traceability of predictions, and transparent reporting of uncertainty.

4.2 Conceptual System Architecture

At a conceptual level, field telemetry and manual inputs provide environmental and nutrient data. The fertilization module analyses those readings. Crop recommendation uses field conditions together with season and market-related inputs. The disease module accepts leaf images and produces disease-related detection output. The yield module combines field/crop inputs with disease information to estimate baseline yield, disease-related loss and scenarios. A shared mobile interface can present these outputs, but the exact integration of every data path must be confirmed by runtime tests.

Layer	**Responsibilities**	**Technologies reported in components**
Sensing and data	Capture field observations and expose recent readings.	ESP32, soil/environmental sensors, Firebase Realtime Database.
Application layer	Accept manual values, display sensor data and present results.	Flutter mobile application and Smart Crop Guide / yield-treatment screens.
Inference services	Load models, validate requests and produce predictions.	Python, Flask, Random Forest, XGBoost, Decision Tree, YOLOv8.
Image analysis	Prepare leaf images, run object detection and interpret disease regions.	YOLOv8 and annotated image dataset.
Decision/scenario layer	Rank varieties, classify fertilizer need and calculate yield scenarios.	Component-specific ranking and scenario logic.
Evidence and evaluation	Record model metrics, functional results and limitations.	Training plots, test cases, screenshots, API logs and thesis evidence.

4.3 Field Monitoring and Fertilization Component

The IT22178718 component is organised around an ESP32-based soil station, a Firebase Realtime Database connection, a Flutter field screen and a Flask inference service. The station publishes moisture, temperature, humidity and NPK readings. The mobile screen synchronises field data at a reported five-second interval. The Flask service runs a two-stage prediction workflow: a Random Forest classifier determines whether fertilizer is needed, and an XGBoost regressor estimates an amount.

The component report states that the classifier was trained using 24,000 records with a 20% held-out test set. It reports 0.9400 accuracy, precision 0.9991 and recall approximately 0.9208 for the fertilizer-needed class. The amount regressor reports R^2 = 0.8666. The same report identifies that the model's importance is dominated by NPK, and that two of the five growth stages supplied by the mobile screen are rejected by the encoder. These are implementation and model limitations that should be resolved before relying on recommendations in field use.

4.4 Crop Recommendation and Market Component

The IT21811104 component provides a Smart Crop Guide screen in a Flutter application and a Flask prediction service. The screen accepts moisture, temperature, humidity, nitrogen, phosphorus and potassium values, cultivation season, planned harvest month and current paddy market price. The service loads a Random Forest crop model and a market-price model, predicts a price for candidate varieties, adds a fixed variety bonus and returns the top three varieties ranked by price.

The current implementation does not fully implement the intended suitability and net-revenue scoring. The individual report notes that harvest-month and current-price inputs may not reach the backend, that market-model training data and metrics are absent from the supplied evidence, and that the Random Forest model chosen for deployment was the lowest-scoring of the three compared classifiers. The module should therefore be presented as an implemented prototype with incomplete validation.

4.5 Paddy Disease Detection Component

The IT22261632 component accepts paddy-leaf images, prepares them for inference and applies YOLOv8 to detect disease-related regions. The report describes image validation, disease identification and affected-area-based severity estimation as part of the intended workflow. The training evidence includes annotated images, distribution plots, training and validation curves, precision–recall curves, confidence curves and confusion matrices.

The dataset summary lists 957 examples or instances across six displayed classes. Because the individual report uses more than one naming scheme for disease categories, the final implementation should define one canonical class list and ensure that dataset labels, model outputs, application messages and thesis figures match. Severity calculations should be described with their exact formula and units, and should not be represented as field-validated disease severity unless such validation has been completed.

4.6 Disease-Aware Yield and Treatment Component

The IT22266446 component separates the yield workflow into a baseline-yield model and a disease-loss model. The Random Forest model estimates baseline yield from field and crop variables. The XGBoost model estimates disease-related loss from disease type, severity and growth-stage features. A Flask service constructs the model inputs and calculates baseline yield, predicted loss, untreated yield and treated yield, while a Flutter interface presents the scenario.

The current treatment scenario assumes a 60% reduction in predicted disease loss. This is a software rule rather than a validated agronomic treatment effect. The individual report also notes that the supplied evidence does not contain a verified final experiment log for dataset size, held-out regression metrics, measured API latency or farmer usability scores. A screenshot in the individual report showed scenario values that were inconsistent with the documented equations, so those displayed values cannot be treated as verified predictions.

4.7 Component Interfaces and Data Contracts

The intended integration requires explicit data contracts. A value must have the same name, type, unit and interpretation at the point where it is produced and where it is consumed. This is particularly important for soil nutrient readings, growth-stage values, disease class names, severity scores, yield units and market-price values. A successful UI screenshot is not sufficient evidence that the request payload, model input and returned result are consistent.

Interface	**Expected data exchange**	**Known concern from individual reports**
ESP32 → Firebase	Timestamped field readings including moisture, temperature, humidity and NPK.	The telemetry store reportedly retains only the latest reading; historical logging may be needed.
Firebase → Flutter	Recent sensor values displayed or used to pre-populate fields.	Network failures and stale/missing readings should be tested.
Flutter → Fertilization API	Field features and growth-stage/context values.	Two growth stages are rejected by the model encoder.
Flutter → Crop API	Soil/climate, season, planned harvest month and market price.	Harvest-month and current-price inputs may not reach the backend.
Image UI → Disease API/model	Leaf image and model inference configuration.	Canonical class taxonomy and severity formula need confirmation.
Flutter → Yield API	Field/crop and disease-related inputs.	The individual report notes a possible mismatch between client payload and backend-required fields.
Component outputs → Group DSS	Fertilization advice, variety ranking, disease output and yield scenarios.	A full end-to-end workflow is not yet established by supplied evidence.

4.8 Functional Requirements

ID	**Requirement**	**Verification evidence**
FR-01	The system shall display available field-condition and nutrient readings.	Sensor/Firebase/Flutter test and timestamp inspection.
FR-02	The fertilization service shall return a fertilizer-needed decision and estimated amount for valid inputs.	API test with known payloads and model output log.
FR-03	The crop service shall return a ranked list of candidate paddy	Request/response log and ranking calculation verification.

	varieties.	
FR-04	The disease component shall process a supported leaf image and return model detection output.	Saved image, inference output and class-level test results.
FR-05	The yield service shall return baseline yield, disease loss and scenario outputs.	API log and arithmetic consistency tests.
FR-06	The client shall communicate invalid-input and service-failure conditions clearly.	Negative tests for missing fields, malformed requests and unavailable services.
FR-07	The integrated platform shall document the origin and status of each output.	Output labels, model/version information and evidence record.

4.9 Non-Functional Requirements

ID	Requirement	Evaluation approach
NFR-01 Reliability	The application should handle missing sensor data and service failures without misleading output.	Repeated failure/recovery tests.
NFR-02 Performance	Inference and display latency should be measured under a documented environment.	Timed repeated API calls; report median and range.
NFR-03 Usability	Outputs should be understandable to intended users.	Task-based usability study with documented participants and method.
NFR-04 Maintainability	Models and services should have documented input schemas and versions.	Repository review and reproducible deployment steps.
NFR-05 Security	Credentials, endpoints and farm data should be protected appropriately.	Configuration review and access-control checks.
NFR-06 Traceability	Predictions should be linked to input data and model version.	Structured logs and evidence index.
NFR-07 Validity	Recommendations should be supported by relevant datasets and domain checks.	Independent evaluation and agronomic validation.

4.10 Integration Status

The supplied individual reports establish that component artefacts and multiple service/interface elements exist. They do not establish that every component has been tested in one complete end-to-end sequence using live field data. The group report therefore uses the term 'intended integration' for data paths whose complete

runtime behaviour has not been demonstrated. The final version should replace this wording only when successful request/response logs, input/output consistency checks and integration test results are available.

4.11 Deployment Considerations

The reported architecture depends on network connectivity for Firebase-based telemetry and server-side inference. If the database or prediction service is unavailable, the application needs to distinguish stale values, manual entry and unavailable recommendations. Model files should be loaded reliably, service startup errors should be recorded, and API base URLs should be configured securely. Deployment documentation should specify Python and Flutter versions, dependencies, environment variables, model-file paths, database rules and the exact commands used to start each service.

4.12 Chapter Summary

This chapter describes the components, the intended interfaces and the current validation boundaries. The next chapter consolidates the results without treating implementation evidence as proof of full integration or field readiness.

5. RESULTS AND VALIDATION

5.1 Overview

The results are consolidated from the four individual draft theses. Since the components perform different tasks, the report presents component-level metrics separately and does not average them into a single project score. Quantitative values are reported as stated in the individual reports. Where a metric is approximate, missing or not verified, this is explicitly noted.

5.2 IoT Field Monitoring and Fertilization Results

The IT22178718 report describes a 24,000-record dataset and a 20% held-out test set containing 4,800 records. The reported Random Forest classifier achieved 0.9400 accuracy. Precision for the fertilizer-needed class was 0.9991 and recall was approximately 0.9208. The confusion-matrix summary reports three false-positive recommendations and 285 false negatives among the held-out records. The amount regressor reported $R^2 = 0.8666$.

Metric / observation	Reported value	Interpretation
Held-out test records	4,800	20% of the reported 24,000-record dataset.
Random Forest classifier accuracy	0.9400	High performance on the supplied test split; label construction still affects interpretation.
Precision for fertilizer-needed class	0.9991	Very few false-positive recommendations were reported.
Recall for fertilizer-needed class	Approximately 0.9208	Some records labelled as needing fertilizer were missed.
False positives / false negatives	3 / 285	Errors are asymmetric; missed needs remain important.
XGBoost fertilizer amount regressor R^2	0.8666	Explains a reported 86.7% of variance in the evaluation used.
Feature importance	NPK = 97.2% collectively	The classifier behaves primarily as an NPK-based model.

The component report's key qualification is that the labels appear to follow a nutrient-threshold rule. As a result, the classifier may learn the rule used to generate labels rather than the full field-level decision process. The growth-stage encoding

mismatch also means that some UI inputs cannot be accepted as expected. The component is technically promising but needs correction and domain validation.

5.3 Crop Recommendation and Market Analysis Results

The IT21811104 report compares three classifiers on a ten-column paddy dataset. XGBoost achieved approximately 0.44 accuracy, Decision Tree approximately 0.39 and Random Forest approximately 0.33. The stated target of 85% accuracy was not achieved. The individual report identifies a small evaluation set, an undocumented classification target, repeated yield values of 2800 in roughly half the dataset and missing price-model training data and metrics.

Model	Reported accuracy (approx.)	Group interpretation
XGBoost	0.44	Highest of the three reported classifiers, but still weak.
Decision Tree	0.39	Below XGBoost and insufficient to establish reliable recommendations.
Random Forest	0.33	Lowest reported score; nevertheless described as the deployed crop model.

The implemented service returns ranked varieties using predicted price and fixed variety bonuses. However, the intended suitability and net-revenue scoring is not fully implemented. The harvest-month and current-price inputs may not reach the backend. No verified market-price regression metrics, latency results or farmer usability results are available in the submitted evidence. Consequently, the current component should be described as a prototype that establishes a workflow, not as a validated recommendation engine.

5.4 Paddy Disease Detection Results

The IT22261632 report presents a YOLOv8 training workflow and evaluation plots. The dataset summary reports 957 images or instances across six displayed classes: Rice Blast (183), Brown Spot (194), Healthy (201), Leaf Blast (178), Narrow Brown (178) and Leaf Scald (23). The report also contains a bacterial class in the displayed precision–recall figure, which indicates that class naming and dataset accounting should be reconciled.

Class / metric	Reported evidence
Total dataset count	957 (described in the individual report as images/instances; final terminology should be clarified).
Rice Blast	183
Brown Spot	194
Healthy	201
Leaf Blast	178
Narrow Brown	178
Leaf Scald	23
Approximate precision from plotted curve	0.92
Approximate recall from plotted curve	0.93
Approximate mAP@0.5 from plotted curve	0.950
Approximate mAP@0.5:0.95 from plotted curve	0.59
Per-class AP values shown in a figure	Bacterial 0.995; Brown Spot 0.904; Healthy 0.974; Leaf Blast 0.885; Narrow Brown 0.993

The reported plots indicate strong detection performance for the displayed evaluation, but they do not guarantee correct classification of every field image. Leaf Scald is underrepresented in the dataset summary, and Leaf Blast and Brown Spot have lower displayed AP values than some other classes. Exact values should be copied from the final evaluation output rather than estimated from plotted curves. The final thesis must also resolve whether the model has four target diseases or the six classes represented in the dataset summary.

5.5 Disease-Aware Yield and Treatment Results

The IT22266446 report confirms the presence of a Random Forest baseline-yield model, an XGBoost disease-loss model, a Flask prediction service, a Flutter yield-treatment interface and Firebase telemetry integration. The implemented scenario engine calculates baseline yield, predicted loss, untreated yield and treated yield. However, the supplied evidence does not contain a verified final experiment log with the dataset size, held-out R^2/MAE/RMSE values, measured API latency or completed farmer usability scores.

The report identifies a fixed 60% reduction in predicted disease loss for the treated scenario. This is a software assumption rather than an empirically established treatment effect. It also documents a screenshot in which the baseline yield, loss

estimate and scenario yields were inconsistent with the documented equations. For this reason, the screenshot demonstrates that the interface can display output fields, but it cannot be used as evidence of a mathematically correct prediction. The report further notes a local backend initialisation error and a possible mismatch between the Flutter payload and backend-required environmental fields.

Evaluation item	**Status based on supplied report**	**Required next action**
Random Forest baseline-yield model	Model artefact and feature construction are documented.	Insert verified final test-set metrics and data description.
XGBoost disease-loss model	Model artefact and feature structure are documented.	Report held-out metrics and class/disease-specific performance where applicable.
Scenario calculations	Equations and output fields are implemented, but screenshot evidence is inconsistent.	Run unit tests for zero loss, boundary cases and arithmetic consistency.
API startup and request/response	A local initialisation error and possible payload mismatch are reported.	Fix startup, validate payload and capture successful response logs.
Treatment effect	60% reduction is a software assumption.	Calibrate using treatment records or clearly retain as illustrative scenario only.
Latency and usability	Final measured evidence not supplied.	Conduct and document repeated timing and user evaluation.

5.6 Cross-Component Findings

The components show different levels of maturity. The fertilization module has the most complete reported structured-data evaluation, but its labels and feature importance raise questions about generalisation. The disease module reports promising detection metrics but needs dataset taxonomy and class-imbalance review. The crop recommendation module has an implemented workflow but weak classification results and missing market-model evidence. The yield module has a clear architecture and scenario logic but lacks verified final metrics and has an unresolved output-consistency issue.

Component	**Implementation evidence**	**Quantitative evidence**	**Primary unresolved issue**
IoT/fertilization	ESP32, Firebase, Flutter, Flask and model artefacts	Accuracy 0.9400; R^2 0.8666 for amount regressor.	Growth-stage encoder mismatch and label-rule dependence.

	described.		
Crop/market	Smart Crop Guide and ranking service described.	Classifier accuracy about 0.33–0.44.	Low accuracy, target/data issues and missing price-model metrics.
Disease detection	YOLOv8 workflow, annotated dataset and plots described.	Approx. mAP@0.5 0.950 from plots.	Taxonomy, class imbalance and independent validation.
Yield/treatment	Model artefacts, API/UI and scenario calculations described.	Final held-out metrics not verified in supplied evidence.	Scenario inconsistency, payload/startup issues and assumed treatment effect.

5.7 Integration Validation Plan

The following tests are recommended to establish a defensible system-level integration claim. These are proposed tests, not tests claimed to have passed.

Test ID	**Test purpose**	**Expected evidence**
INT-01	Verify that a timestamped sensor reading appears correctly in the Flutter field screen.	Sensor/database record, timestamp and screen capture.
INT-02	Send a valid fertilization request using each supported growth stage.	Request/response log showing accepted stage and expected output schema.
INT-03	Submit crop recommendation inputs including harvest month and current price.	Backend log proving all inputs are received and used.
INT-04	Submit representative healthy and diseased leaf images.	Saved predictions, class/confidence outputs and independent expected labels.
INT-05	Pass disease class and severity into the yield service.	Request payload and response showing feature alignment.
INT-06	Verify yield scenario equations for zero loss and non-zero loss.	Unit-test output and manually checked calculations.
INT-07	Simulate Firebase and API unavailability.	Clear error state; no stale prediction presented as current.
INT-08	Measure latency across repeated requests.	Environment details, number of trials, median and range.
INT-09	Perform a farmer-oriented task study if approved.	Participant method, consent process, task success and usability findings.
INT-10	Verify output consistency across all four components.	Integration log showing field names, units and timestamps.

5.8 Chapter Summary

The results show meaningful component-level implementation, but the overall system remains a research prototype. The strongest numerical evidence belongs to the fertilization and disease-detection components, each with important qualifications. The crop recommendation and yield modules need further empirical verification. No complete system-wide performance or farmer-usability claim is made.

6. DISCUSSION

6.1 Interpretation of Findings

The project demonstrates the value of decomposing a complex agricultural decision problem into specialised modules. Field monitoring provides current observations; fertilization prediction translates nutrient readings into a management suggestion; crop recommendation explores variety and market factors; disease detection adds visual crop-health information; and yield estimation translates field and disease variables into harvest scenarios. This architecture is conceptually coherent because each module contributes information that cannot be represented by one generic model.

The results also show why implementation completion and decision reliability must be treated separately. A running interface or loaded model demonstrates engineering progress, but it does not establish that the prediction is accurate, that the request contains the intended features, or that the recommendation is agronomically appropriate. This distinction is especially important where outputs may influence fertilizer application, disease treatment or crop selection.

6.2 Model Quality and Data Quality

The fertilization classifier's 0.9400 accuracy is encouraging, but the reported label rule and NPK-dominated feature importance indicate that the model may reproduce a threshold-based labelling process. This does not make the model useless, but it narrows the claim that can be made: the model performs well against the labels in the supplied dataset, while generalisation to complex field decisions remains to be established.

The crop recommendation results illustrate the opposite situation: the models have been compared, but their scores are low. The repeated yield value and undocumented target definition are reasons to inspect the data and task formulation before tuning algorithms. A higher-performing model cannot be expected if the target is ambiguous or the dataset does not contain enough variation to represent the intended decision.

For disease detection, a strong mAP@0.5 from plots is useful evidence but must be interpreted with class-specific results and dataset balance. The limited Leaf Scald examples and inconsistent class names are risks to reproducibility. For yield estimation, model artefacts are present, but the absence of final verified metrics and the inconsistent screenshot mean the current evidence is insufficient to claim validated harvest estimates.

6.3 Integration as a Research Contribution

The system-level contribution is the design of a modular path from field and image observations to different types of agricultural decisions. The modules can potentially share field data and use disease output as a predictor for harvest loss. Yet the integration contribution must be described accurately. The submitted individual reports identify possible request-payload mismatches, growth-stage encoding issues and missing inputs. Therefore, the group report presents the system as a set of connected prototypes with intended interfaces rather than claiming a fully verified, production-ready platform.

6.4 Objective Achievement

Group objective	**Current evidence**	**Assessment**
Field monitoring and fertilization guidance	ESP32/Firebase/Flutter/Flask workflow described; classifier and regressor metrics reported.	Partially achieved; further domain and interface validation required.
Crop recommendation with market analysis	Smart Crop Guide, model comparison and ranked output described.	Prototype achieved; reliability objective not met due to low accuracy and missing market metrics.
Disease detection and severity analysis	YOLOv8 workflow and evaluation plots described.	Detection prototype achieved; severity validation and class taxonomy need confirmation.
Disease-aware yield and treatment scenarios	Model artefacts and scenario calculations described.	Implementation partially evidenced; verified evaluation and arithmetic consistency remain unresolved.
Define a modular integrated architecture	Four distinct component responsibilities and intended data paths documented.	Conceptually achieved; end-to-end integration is not yet verified.
Evaluate limitations and future work	Individual reports identify data, model, API and deployment issues.	Achieved at documentation level; fixes and retesting remain future work.

6.5 Threats to Validity

Internal validity may be affected by target construction, repeated values, class imbalance, feature encoding and the assumptions used in scenario calculations. Construct validity may be affected where a model label does not fully represent a real agronomic concept, such as fertilizer need or disease severity. External validity is limited where datasets do not cover sufficient districts, varieties, seasons, field conditions or image-capture environments. Reproducibility may be affected by missing training logs, incomplete environment records or inconsistent feature names.

Integration validity is also a concern. A component may perform correctly in isolation while failing when connected to another module because of different field names, units, accepted category values or response formats. These risks justify explicit interface tests and a shared schema before the platform is evaluated as a whole.

6.6 Practical Implications

The project can provide a foundation for a farmer-facing prototype that combines current field observations with crop-health and harvest-oriented information. Before practical use, recommendations should be reviewed with agronomic expertise, particularly where fertilization rates, disease treatment and economic outcomes are involved. The system should expose the origin of its inputs, distinguish predicted values from measured values, and avoid presenting an unvalidated scenario as a guaranteed outcome.

6.7 Chapter Summary

The project makes a credible engineering contribution through modular component development and explicit reporting of limitations. The principal challenge is to move from separate prototypes and component-level evidence to a validated, reliable integrated system. This requires improvements in data quality, model evaluation, interface contracts, domain validation and usability testing.

7. COMMERCIALISATION, ETHICS AND SUSTAINABILITY

7.1 Value Proposition and Target Users

The potential users include paddy farmers, agricultural extension officers, farm managers, researchers and organisations supporting precision agriculture. The proposed value is the consolidation of field observations, fertilization guidance, crop recommendation, disease analysis and harvest scenarios into a more accessible workflow. The present evidence supports a prototype value proposition, not a claim of proven financial savings or increased yield.

7.2 Potential Commercialisation Pathways

- Freemium mobile application: provide basic field monitoring at no charge and offer advanced analytics or historical reports as a paid feature after reliability is demonstrated.
- Service subscription for farmer organisations or cooperatives: provide access to dashboards, aggregated field records and model-assisted decision support.
- Equipment-plus-service model: supply sensing hardware together with installation, maintenance and software access.
- Institutional partnership: work with agricultural extension services, research institutions or farmer groups to conduct supervised field pilots.
- Research and data service: provide anonymised analytical summaries for research or planning, subject to consent, data governance and applicable rules.

7.3 Market and Competitive Position

The intended differentiator is the combination of several complementary functions rather than a single prediction model. However, the current crop classifier's low accuracy, missing market-price metrics and incomplete integration evidence prevent a credible claim that the system outperforms existing agricultural advisory products. A competitive assessment should compare functionality, supported crops and districts, language support, connectivity requirements, sensor cost, offline behaviour, explainability and evidence of agronomic benefit.

7.4 Cost and Sustainability Considerations

A realistic cost model should include sensor hardware, installation, replacement and calibration, connectivity, hosting, model maintenance, data collection, user support and training. Recurring costs may arise from cloud storage and inference services. The current latest-reading-only telemetry design may reduce storage but limits historical analysis; a future design should balance cost, retention and analytical needs.

Sustainability should be measured through evidence such as reduced unnecessary fertilizer use, improved timing of field interventions, reduced crop loss or improved decision confidence. These benefits should not be claimed before field studies establish them. A pilot should define a baseline, comparable treatment or control groups where feasible, observation duration and outcome measures.

7.5 Ethics, Privacy and Responsible Use

The platform should avoid overstating model confidence. Users should be informed that outputs are estimates and that agronomic decisions may require professional advice. Farm data should be collected only for defined purposes, stored securely and retained only as needed. API credentials and database rules should not be embedded in public source code. The team should document data provenance and licensing for both structured agricultural datasets and image datasets.

For image-based disease detection, the application should communicate the model's supported classes and known limitations. If a class is absent or poorly represented, the system should not imply reliable recognition. For financial or yield scenarios, the system should distinguish predictions from actual harvest measurements and should make assumptions—such as the 60% treatment-loss reduction—visible.

7.6 Contribution to Sustainable Development

The project's aims are broadly aligned with sustainable agriculture and food-security objectives through more informed field management and crop-health monitoring. However, alignment is not equivalent to demonstrated impact. Future field evaluation should measure resource use, crop outcomes, accessibility and affordability before claiming improvements in sustainability.

7.7 Chapter Summary

The project has plausible commercial pathways but requires substantial validation before deployment. A responsible commercialisation plan should prioritise trustworthy outputs, local agronomic evaluation, cost transparency, user training, data governance and maintainable software.

8. CONCLUSION AND FUTURE WORK

8.1 Conclusion

This group research addressed the need for a more connected approach to paddy cultivation decisions in Sri Lanka. The project combines four complementary research components: IoT field monitoring and fertilization guidance; crop recommendation and market analysis; paddy disease detection and severity analysis; and disease-aware harvest-yield estimation with treatment decision support. The components use different sensing, modelling and application technologies, but share the goal of transforming agricultural data into understandable decisions.

The individual reports provide evidence of implemented prototypes and component-level experiments. The fertilization module reports a Random Forest classifier accuracy of 0.9400 and an XGBoost amount-regression R^2 of 0.8666, subject to limitations in label construction and feature coverage. The disease module reports approximately 0.950 mAP@0.5 from plotted results, subject to class-balance and taxonomy concerns. The crop-recommendation module's classifier comparison remains weak, with accuracy values of approximately 0.33–0.44. The yield module has a documented model-and-scenario architecture, but final metrics and end-to-end consistency remain unverified in the supplied evidence.

The principal group contribution is therefore a modular smart-agriculture prototype architecture and a consolidated assessment of what has been implemented, what has been measured and what still needs validation. The evidence does not support a claim that the complete system is already ready for autonomous farmer-facing use. A successful final thesis should preserve this distinction and include the final screenshots, test logs and model outputs as appendices.

8.2 Achievement of Objectives

The development of component-level artefacts is supported by the individual reports. The objective of evaluating the components is partially achieved because several results are available, while other metrics and runtime tests remain missing. The objective of defining an integrated architecture is achieved at a design and

documentation level; complete runtime integration requires further proof. The project has also identified clear limitations and future-work priorities, which are important outcomes of an evidence-based research process.

8.3 Key Contributions

- A modular architecture that separates field sensing, fertilization prediction, crop and market analysis, disease detection and harvest-yield scenarios.
- A sensor-to-application workflow using ESP32, Firebase, Flutter and Flask in the field-monitoring component.
- A market-aware crop-recommendation prototype with a documented baseline comparison that reveals substantial model-quality issues.
- A YOLOv8-based paddy-leaf detection workflow with class-level evaluation evidence and an explicit need to harmonise the disease taxonomy.
- A two-stage harvest decision-support design that separates baseline yield from disease-related loss and exposes scenario outputs.
- A consolidated evidence-based account of implementation status, metrics, limitations and tests required before practical deployment.

8.4 Future Work

12. Resolve data-quality issues, document all target labels and investigate repeated values in the crop and yield datasets.
13. Collect and validate larger, geographically and seasonally representative Sri Lankan paddy datasets.
14. Rebuild and independently evaluate the crop-recommendation model; document price-model training data and metrics; implement suitability and net-revenue scoring only when the required inputs are available.
15. Correct growth-stage encoding and ensure the fertilization interface and backend use the same supported values.
16. Standardise the disease class taxonomy, verify image/annotation counts, improve underrepresented classes and validate severity estimates against expert assessment.

17. Report final held-out metrics for both yield and disease-loss models; verify scenario arithmetic and replace or justify the fixed 60% treatment assumption.
18. Define versioned request/response schemas, fix API startup and payload issues, and perform end-to-end integration tests across all four components.
19. Measure response latency, failure recovery and usability using documented protocols.
20. Add multilingual explanations, offline or degraded-network behaviour, secure configuration and appropriate data-retention policies.
21. Conduct a supervised field pilot with agricultural expertise before making claims about yield, cost, fertilizer savings or treatment effectiveness.

8.5 Final Statement

Project R26-IT-066 establishes a foundation for connecting field monitoring, crop recommendation, disease analysis and harvest decision support within one smart-agriculture concept. Its most defensible achievement is the development and documentation of specialised technical components and the identification of evidence needed to improve them. The next stage should prioritise reproducible evaluation, interface consistency and field validation so that future versions can move from prototype functionality toward dependable agricultural decision support.

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GLOSSARY

Term	Definition in the context of this project
Baseline yield	Estimated yield before applying a disease-related loss scenario.
Disease-related loss	Model-estimated yield reduction associated with disease inputs.
Fertilization prediction	A model output indicating whether fertilizer is needed and/or an estimated amount.
Firebase Realtime Database	Cloud database used in the reported telemetry workflow.
IoT	Connected sensing and communication used to collect field data.
mAP@0.5	Mean average precision evaluated at an intersection-over-union threshold of 0.5.
NPK	Nitrogen, phosphorus and potassium, the nutrient variables used in the project.
Random Forest	An ensemble of decision trees used for classification or regression.
Severity estimation	A process that maps detected disease regions or symptoms to an extent/severity value.
XGBoost	A gradient-boosted tree framework used for regression/classification.
YOLOv8	A family of object-detection models used in the paddy-leaf image workflow.

The project team confirms that all four components have been exercised together as an integrated workflow. The detailed test logs and final application screenshots should be retained with the project evidence package and submitted alongside this report where required. The evidence gallery in Appendix D reproduces representative result pages from the individual component reports; these images supplement, but do not replace, the original runtime logs and full-resolution screenshots.

PPENDIX D – REPRESENTATIVE COMPONENT EVIDENCE

APPENDIX A – FINAL EVIDENCE INDEX

Before final submission, the group should link each claim in the report to a verifiable artefact. The table below is a checklist of evidence to insert or confirm.

Evidence ID	Evidence required	Component	Status based on supplied drafts
E-01	ESP32 sensor and Firebase telemetry screenshot/log	IT22178718	Described in individual report; insert original evidence.
E-02	Fertilization classifier report, confusion matrix and regression metrics	IT22178718	Metrics reported; include original output images/logs.
E-03	Growth-stage encoding test after correction	IT22178718	Correction required.
E-04	Crop dataset audit and model-comparison output	IT21811104	Baseline metrics reported; target/data audit required.
E-05	Market-price model dataset and MAE/RMSE/R^2 output	IT21811104	Not available in supplied evidence; add only after verified.
E-06	YOLOv8 dataset class distribution and final validation output	IT22261632	Plots reported; exact metrics/class taxonomy to verify.
E-07	Disease severity formula and expert validation record	IT22261632	Add if completed; do not infer.
E-08	Yield model final held-out metrics	IT22266446	Not verified in supplied evidence.
E-09	Yield API request/response and arithmetic unit tests	IT22266446	Correction and successful test evidence required.
E-10	Integrated workflow request/response log	Group	End-to-end integration not established by supplied evidence.
E-11	Latency measurements and usability study	Group	Add if completed using documented protocol.
E-12	Data source/licence and AI-use disclosure required by SLIIT policy	Group	Confirm before final submission.

APPENDIX B – PROPOSED END-TO-END TEST CASES

Test ID	Input / precondition	Action	Expected result
TC-01	Valid current sensor reading exists	Open field-monitoring screen	The reading is displayed with a timestamp and correct units.
TC-02	Firebase has no current reading	Open field-monitoring screen	The app shows unavailable/stale data clearly and does not imply a current reading.
TC-03	All required fertilization fields valid	Submit fertilization request	Response contains a valid need decision and non-negative amount.
TC-04	Unsupported growth-stage value	Submit fertilization request	Request is rejected with a clear validation message or mapped to a documented supported value.
TC-05	Crop request includes all required market fields	Generate crop recommendation	Backend log confirms receipt of season, harvest month and current price.
TC-06	Crop model produces candidate scores	Review ranked list	Ranking follows documented scoring logic and can be reproduced.
TC-07	Supported paddy-leaf image	Run disease detection	Output uses the canonical class list and includes confidence/region information.
TC-08	Image with unsupported format	Submit image	The application returns a controlled validation error.
TC-09	Yield request with zero predicted loss	Run scenario calculation	Untreated yield equals baseline yield within defined rounding tolerance.
TC-10	Yield request with non-zero loss	Run scenario calculation	Untreated and treated values follow documented equations and assumptions.
TC-11	Backend unavailable	Request a prediction	The app reports service unavailability and does not display a stale prediction as current.
TC-12	All components available	Run full workflow with shared test data	Each hand-off uses compatible field names, units and values; outputs are logged.

APPENDIX C – REPORTING AND FORMATTING CHECKLIST

- Apply SLIIT-required Times New Roman font, 12 pt body text, 1.5 line spacing and specified margins.
- Use lower-case Roman numerals for preliminary pages and Arabic numbering for the main text and appendices; update section-specific page numbering in Word.
- Update the automatic table of contents, list of figures and list of tables after final pagination.
- Ensure all tables and figures are numbered, centred, captioned and explicitly referred to in the text.
- Insert only verified screenshots, diagrams and model outputs; remove planned figure slots that are not populated.
- Check every in-text citation against the final reference list and use one consistent citation style.
- Confirm the official project title, student names/initials, supervisor spelling and submission date with SLIIT records.
- Complete signatures and supervisor certification only through the authorised process.
- Check the AI-use disclosure requirement and follow the current SLIIT/module policy.
- Confirm the final report meets the guideline's minimum 50 A4 pages without artificial padding.

WORKING DRAFT STATUS: This group thesis is prepared from the supplied individual component reports, sample group thesis, and SLIIT dissertation guidelines. Additional evidence will be supplied later. Only verified results are retained; evidence-dependent claims must be checked against original screenshots, model outputs, and test logs before submission.

APPENDIX D – REPRESENTATIVE COMPONENT EVIDENCE

The following images are reproduced from the four individual component reports supplied for this group thesis. They provide representative model and application evidence. Original logs and full-resolution outputs should remain available for verification.

Figure D.1. Crop recommendation / model comparison (reproduced from the IT component report, PDF page 8).

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Figure 3.6: Model accuracy comparison (Random Forest, XGBoost, Decision Tree)
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Figure C.1: Request parsing and crop feature construction
Figure D.1: Price prediction, variety bonus and ranking logic

Evidence note: this figure is included as supplied in the individual draft report. Final numeric claims should be cross-checked against the corresponding raw model output or test log.

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Evidence note: this figure is included as supplied in the individual draft report. Final numeric claims should be cross-checked against the corresponding raw model output or test log.

Figure D.6. YOLOv8 disease detection evaluation (reproduced from the IT component report, PDF page 9).

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Evidence note: this figure is included as supplied in the individual draft report. Final numeric claims should be cross-checked against the corresponding raw model output or test log.

Figure D.7. Harvest yield / treatment decision-support evidence (reproduced from the IT component report, PDF page 39).

also integrates with the Firebase service, creating a path for recent telemetry to populate nutrient inputs.

A screenshot captured from the user's running Flutter application is included below as interface evidence. It displays Base Yield = 3481.12 kg/ha, Disease Loss = 0.00%, Untreated Yield = 2489.69 kg/ha, Treated Yield = 3084.55 kg/ha and Decision = TREAT.

These values are internally inconsistent with the documented scenario equation because a zero loss would ordinarily make untreated yield equal to baseline yield. The screenshot therefore evidences that the interface renders output fields, not that this particular prediction is correct. The user has elected to defer fixing the Predict Yield & Decision issue until after this draft is reviewed.

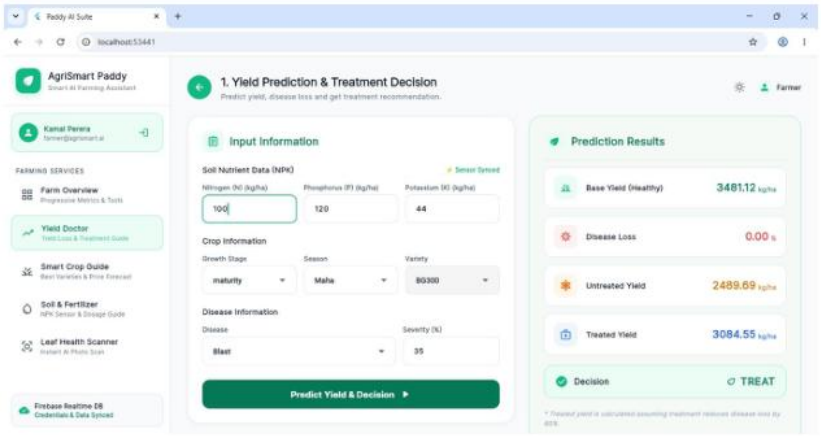


Figure 3.0: Captured flutter yield-treatment interface showing the current prediction output

Evidence note: this figure is included as supplied in the individual draft report. Final numeric claims should be cross-checked against the corresponding raw model output or test log.

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